

ECO 362 Financial Investments

Final Exam

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Fall 2022

Instructions

1. Check that you have all 5 pages. There are 6 questions.

Question	Points
1	14
2	20
3	16
4	16
5	16
6	18
Total	100

2. This is a closed book exam. However, you may use a two-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on both sides. You are under the Honor Code to comply with this requirement.
3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.
4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book:
I pledge my honor that I have not violated the Honor Code during this examination.
5. Write your answers in the exam books. *Show all steps necessary in deriving your solution for full credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., \$95.67 or 1.23%).
6. You have 3 hours to complete the exam. After 3 hours, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.
7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. Answer TRUE or FALSE. No explanation is necessary.

[14 points: 2 points each part]

- (a) An investor with mean-variance expected utility should never hold a stock that has expected return lower than the riskless interest rate.
- (b) According to the arbitrage pricing theory, stocks with high idiosyncratic risk could earn a negative alpha.
- (c) Actively managed mutual funds have gained market share relative to index funds over the last 15 years.
- (d) The information ratio could be used to compare the performance of fund managers with different benchmark portfolios, such as a small-cap value fund versus a large-cap growth fund.
- (e) Suppose that life insurance is overpriced such that its expected return is lower than the riskless interest rate. You should not buy any life insurance and instead invest in the riskless asset.
- (f) Consider a call option and a put option with the same underlying asset and the same maturity. However, they have different exercise prices. If the Black-Scholes model were true, they could have different implied volatility.
- (g) It is suboptimal to simultaneously hold Treasury bonds in a brokerage account and tax-exempt municipal bonds in a 401(k) account.

2. The following table reports the price-dividend ratio for five stocks.

Stock	P/D
Apple	167
Exxon Mobil	31
Johnson & Johnson	39
Microsoft	93
Walmart	63

According to the Gordon growth model, the price-dividend ratio of stock i is

$$\frac{P_i}{D_i} = \frac{1 + G_i}{\bar{R}_i - G_i},$$

where $\bar{R}_i$ is the expected annual return, and G_i is the expected annual dividend growth.

[20 points: 4 points for each part]

- (a) Of the five stocks, which two are value stocks?
- (b) Of the five stocks, which two are growth stocks?
- (c) Suppose that Exxon Mobil and Johnson & Johnson have the same expected dividend growth $G_i \geq 0$. Which stock has the higher expected return? Explain the reason for your answer.

- (d) Microsoft's expected dividend growth is 5%. What must be Apple's expected dividend growth so that its expected return is the same as Microsoft's expected return?
- (e) Assume no arbitrage. If expected dividend growth is 7% for Apple and 5% for Walmart, which is the riskier stock? Explain the reason for your answer.
3. We discussed the following table in the "Alpha" lecture. It reports the monthly excess returns, standard deviation of monthly returns, market beta, SMB (Small Minus Big) beta, HML (High Minus Low book-to-market equity) beta, WML (Winner Minus Loser) beta, and alpha for 10 portfolios of actively managed mutual funds sorted by past 1-year returns.

Statistic	Portfolio									
	Low	2	3	4	5	6	7	8	9	High
Mean (%)	0.01	0.23	0.34	0.36	0.40	0.38	0.45	0.43	0.59	0.68
SD (%)	4.90	4.60	4.48	4.30	4.36	4.35	4.41	4.56	4.72	5.04
Market beta	0.93	0.93	0.93	0.90	0.90	0.90	0.90	0.90	0.89	0.88
SMB beta	0.32	0.22	0.20	0.21	0.22	0.22	0.27	0.34	0.46	0.62
HML beta	-0.08	-0.10	-0.06	-0.03	-0.04	-0.05	-0.05	-0.07	-0.05	-0.05
WML beta	-0.09	-0.02	0.01	0.04	0.08	0.07	0.11	0.16	0.20	0.29
Alpha (%)	-0.40	-0.20	-0.13	-0.14	-0.12	-0.14	-0.12	-0.18	-0.10	-0.12
Alpha (<i>t</i> -stat)	-4.33	-3.11	-2.52	-3.09	-2.82	-3.31	-2.81	-3.65	-1.78	-1.60

Source: Cahart (1997, *Journal of Finance*)

[16 points: 4 points each part]

- (a) Explain why the alpha is negative for all 10 portfolios.
- (b) The High portfolio (i.e., mutual funds with the best returns over the past year) has the highest return of 0.68% per month. Does this mean that the fund managers in this portfolio have skill? Explain the reason for your answer.
- (c) On average, are actively managed mutual funds small-cap value, small-cap growth, large-cap value, or large cap-growth strategies? Explain the reason for your answer.
- (d) Based on the information given, would you be able to compute average returns on the four factors (market, SMB, HML, and WML)? If no, simply state no for full credit. If yes, write down a system of equations that you could solve for the answer. You do not need to actually solve the system for full credit.
4. The following table reports the continuously compounded zero-coupon Treasury yields in percent.

Maturity	Yield
1	4.57
5	3.87
10	3.66

According to the term structure model, the t -year continuously compounded zero-coupon yield is

$$y(t) = \bar{y} + \frac{1}{t} \left(\frac{1 - \rho^t}{1 - \rho} \right) (y(1) - \bar{y}) + \frac{t-1}{t} \mu,$$

where $\rho = 0.4$. Assume no arbitrage.

[16 points: 4 points for each part]

- Describe what $\bar{y}$ represents. What is the value of $\bar{y}$ that is consistent with the term structure model?
 - Describe what μ represents. What is the value of μ that is consistent with the term structure model?
 - Is the 1-year interest rate expected to rise or fall?
 - What is the continuously compounded zero-coupon yield at the 20-year maturity?
5. The following table reports the continuously compounded zero-coupon yields in percent for nominal Treasuries and TIPS (Treasury inflation protected securities).

Maturity	TIPS	Nominal Treasury
2	2.00	4.35
5	1.48	3.82
10	1.49	3.68

[16 points: 4 points for each part]

- Assume that the inflation risk premium is zero. Does the market think that expected annual inflation will rise, fall, or stay flat over the next 5 years? Explain the reason for your answer. Hint: You need to use the rows for 2- and 5-year maturities in the table above.
- Assume that the inflation risk premium is zero. Does the market think that expected annual inflation will rise, fall, or stay flat over the next 10 years? Explain the reason for your answer. Hint: You need to use the rows for 5- and 10-year maturities in the table above.
- Define the inflation risk premium. Explain how the inflation risk premium could be positive or negative.

(d) Suppose that expected annual inflation over the next 5 years is 4%. That is,

$$\frac{\ln(\mathbb{E}[\text{CPI}_5/\text{CPI}_0])}{5} = 4\%,$$

where CPI_t is the consumer price index in year t . What is the inflation risk premium at the 5-year maturity?

6. In 2004, a university endowment fund enters a 10-year interest rate swap worth \$2 billion in notional amount, receiving a 1-year floating rate and paying a fixed rate. The annually compounded interest rate in year t is used to determine the payment in year $t + 1$. The contract has 10 annual payments in each year from 2005 through 2014. The following table reports the prices of zero-coupon Treasury bonds per \$100 face amount in 2004 and 2009. Assume no arbitrage.

Maturity	1	2	3	4	5	6	7	8	9	10
2004	97.27	94.05	90.65	87.10	83.42	79.64	75.78	71.92	68.10	64.36
2009	99.73	98.72	96.75	93.90	90.37	86.41	82.24	78.04	73.92	69.97

[18 points: 6 points for part (c), 3 points for other parts]

- What is the floating rate on the interest rate swap in 2004 (which determines the payment in 2005)?
- What is the fixed rate on the interest rate swap in 2004?
- What is the market value of the interest rate swap in 2009 (after the payment in 2009)?
- What is the floating rate on the interest rate swap in 2009 (which determines the payment in 2010)?
- What is the net payment from the interest rate swap in 2010? Use a positive number if the university endowment fund receives payment and a negative number otherwise.